

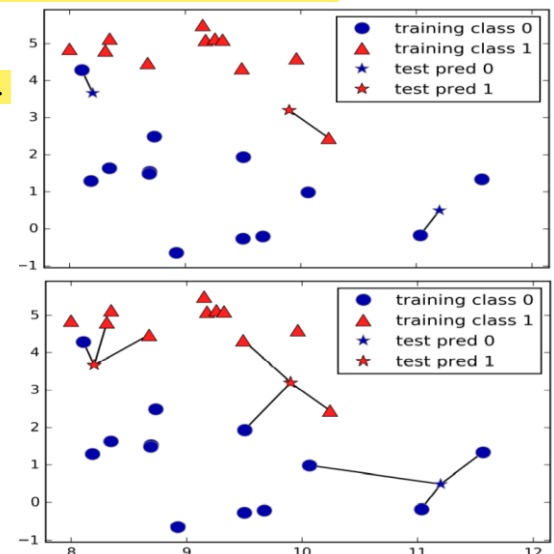
k Nearest Neighbors (KNN)

k-Nearest Neighbor (KNN) Classifier

- The k-Nearest Neighbors classifier (k-NN) is one of the simplest yet most commonly used classifiers in supervised machine learning.
- The basic idea behind KNN is that similar objects are likely to belong to the same class.

Algorithm

- Instead of building an explicit mathematical model during training, KNN simply stores the training examples and delays all computation until a new sample arrives. computation only happens on the arrival of new data point
- When a new object x arrives, the KNN algorithm:
 - Among the training examples, identify the k nearest neighbors of x (examples most similar to x).
 - Let c_i be the class most frequently found among these k nearest neighbors.
 - Label x with c_i
- Because of this behavior, KNN is called a lazy learner, or instance-based learner, or memory-based learner.
- KNN therefore relies heavily on similarity measures, distance calculations, and neighboring examples.
- It searches the m -dimensional feature space for nearby example.
- In its simplest form $k=1$ (1-NN) only the single closest neighbor is used.
- The new sample receives the same class as its nearest neighbor.
- Very simple technique; works well in clean datasets.
- Sensitive to noise, outliers and mislabeled examples.
- Although 1-NN is simple, relying on a single neighbor can be risky in noisy datasets. Also, if the nearest neighbor is incorrectly labeled, then 1-NN may make a wrong prediction.
- Using multiple neighbors helps reduce the effect of noise, outliers and mislabeled training samples.
- This leads to the generalized KNN algorithm.
- Instead of using only one neighbor use k nearest neighbors. The predicted class is determined using majority voting.



Distance Measures in KNN

- To determine similarity, KNN uses distance measures.
- The fact that each example can be represented by a point in an m -dimensional feature space makes it possible to calculate the geometric distance between any pair of examples.
- The closer two examples are in feature space, the greater their similarity.

Euclidean Distance

- Euclidean distance is the straight-line or 'airplane' distance between two points.
- For two points $X = (x_1, x_2, \dots, x_m)$ and $Y = (y_1, y_2, \dots, y_m)$
Euclidean distance, $d(X, Y) = \sqrt{\sum_{i=1}^m (x_i - y_i)^2}$

Manhattan Distance

- Manhattan distance is also called city-block or 'taxicab' distance.
- It measures movement along horizontal and vertical directions.
- For two points $X = (x_1, x_2, \dots, x_m)$ and $Y = (y_1, y_2, \dots, y_m)$
Manhattan distance, $d(X, Y) = \sum_{i=1}^m |x_i - y_i|$

Minkowski Distance

- The Minkowski Distance is a generalized distance measure used in Machine Learning and mathematics.
- It provides a single framework for multiple distance measures.
- For two points $X = (x_1, x_2, \dots, x_m)$ and $Y = (y_1, y_2, \dots, y_m)$

Minkowski distance, $d(X, Y) = (\sum_{i=1}^m |x_i - y_i|^p)^{\frac{1}{p}}$

- p is called the order of the metric.

k-Nearest Neighbor (KNN) Regressor

- Predicts continuous numeric values using averaging

Algorithm

- Compute distances between x and all training examples
- Find the k nearest neighbors
- Take the average of their target values
- KNN regression can produce very noisy predictions and poor extrapolation outside training region because predictions depend entirely on nearby samples.
- It cannot learn a global mathematical trend like linear regression and neural networks.

Weighted Nearest Neighbor

- A KNN model may predict the output of a new sample x based on the majority class or average target value of its nearest neighbors. However, some of these neighbors may be far away from x , while closer neighbors may provide more reliable information. Weighted voting addresses this issue by assigning greater importance to nearer neighbors.
- In weighted voting, the weight assigned to each nearest neighbor is inversely proportional to its distance from the new sample. The closer the neighbor, the greater its impact on the final prediction.
- One commonly used weighting scheme is:
 $w_i = \frac{1}{d(x, x_i)}$ where w_i is the weight assigned to neighbor x_i , and $d(x, x_i)$ is the distance between the new sample x and neighbor x_i .
- The weighted k-NN classifier sums the weights of neighbors supporting the positive class and separately sums the weights of neighbors supporting the negative class.
Weight of class $c = \sum w_i$, for neighbors belonging to class c .
- The final prediction is based on which class receives the larger total weight.
- The weighted k-NN regressor uses the weighted average to make predictions.
 $\hat{y} = \frac{\sum_{i=1}^m w_i y_i}{\sum_{i=1}^m w_i}$, where y_i is the target value of neighbor x_i . $m = k$ (the number of neighbors)

Choosing the Value of k

- The choice of k is an important hyperparameter of the algorithm.
- Small k is sensitive to noise.
- Large k gives smoother decision boundaries and is less sensitive to noise.
- If k becomes too large distant neighbors may dominate and performance may decrease.
- In two-class problems odd values of k are preferred as it reduces the possibility of ties.
- For problems with more than two classes, ties can still occur even with odd k .
- Hence some tie-breaking strategy may be required.
- Possible tie-breaking strategies include choose nearest among tied classes, random selection, weighted voting.
weighted voting is weighted classification

Effect of Data Pre-processing

Irrelevant Attributes

- Irrelevant attributes can reduce classifier performance.
- If only a few attributes are irrelevant impact may be small; if many attributes are irrelevant distance becomes less meaningful, nearest neighbors may no longer be truly similar.

Scaling Problems in KNN

- KNN relies directly on distance calculations. Therefore, attribute scales matter significantly.
- To solve scaling problems the attributes must be normalized or standardized.